

# BST Physics Curriculum Vol 13 Chapter 1 — Biology Substrate Reading v0.3 (Saturday Wave 3)

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2026-05-23 Saturday EDT (Wave 3 second chapter; joint Lyra+Elie)

## Vol 13 Chapter 1 — Biology Substrate Reading

### Chapter motivation

Standard biology: genetic code DNA  $\rightarrow$  mRNA  $\rightarrow$  protein via 4-letter alphabet (A, T, C, G) + 64 codons  $\rightarrow$  20 amino acids + stop. BST identifies the genetic code as substrate-derived from BST primary integers per existing biology track (T452-T467 batches 52-56 + Toys 541-545). Casey's April 2026 insight: "DNA + proton are siblings — each level expresses subset of five integers" — biology is substrate at biological scale.

### Section 1.0b — Reader-grade 3-level pedagogy

Level 1: Genetic code (DNA + 64 codons + 20 amino acids) is substrate-derived from BST primary integers per existing biology track (T452-T467 + Toys 541-545); Casey insight: "DNA + proton are siblings — each level expresses subset of five integers"; biology = substrate at biological scale.

Level 2 (graduate): Standard genetic code: 4 nucleotides (A, T, C, G)  $\rightarrow$  64 codons ( $4^3 = 64$ )  $\rightarrow$  20 amino acids + stop. Standard treatment: code is "frozen accident" or evolved arbitrarily. BST identifies as substrate-derived: per existing biology track (T452-T467 batches 52-56 + Toys 541-545),  $4 = 2^2$  nucleotides (rank<sup>2</sup> match);  $64 = 2^6$  codons ( $2^{\wedge}C\_2$  match); 20 amino acids: substrate-cartography reading via degeneracy patterns + 5-integer BST primary structure. Per Casey insight April 2026: "DNA + proton are siblings — each level expresses subset of five integers"; this is the substrate-cartography framing for biology. Proton substrate-derivation:  $m\_p/m\_e = 6\pi^5$  (Vol 2 Ch 6 CROWN JEWEL T187 at 0.002%); DNA substrate-derivation: genetic-code degeneracy + amino-acid count + codon structure inherit BST primary integers. Substrate genetic code derivation: 64 codons = full 3-nucleotide combination; 20 amino acids = substrate-natural reduction via degeneracy patterns; stop codons = substrate Zone-3 (emission) signature. Prebiotic chemistry +

evolutionary forcing: substrate-natural reduction pathways from primordial soup → first cells per substrate-bias toward BST primary integer patterns.

Level 3 (5th-grader): Biology has a genetic code: 4 DNA letters (A, T, C, G) → 64 three-letter codons → 20 amino acids + stop signals. Standard biology treats this as evolved or accidental. BST identifies the genetic code as substrate-derived from BST primary integers: 4 nucleotides =  $2^2$  (BST rank squared); 64 codons =  $2^6$  (BST  $C_2 = 6$ ); 20 amino acids has substrate-cartography reading per existing biology theorems (T452-T467 + Toys 541-545). Casey April 2026 insight: “DNA + proton are siblings — each level expresses subset of five integers”. Biology is just substrate physics at biological scale.

## Section 1.1 — Standard Genetic Code

DNA: 4 nucleotides (A, T, C, G). Codons:  $4^3 = 64$  (triplet code). Amino acids: 20 + stop codons. Standard view: “frozen accident” or evolved arbitrarily.

## Section 1.2 — BST Substrate-Derived Reading

Per T452-T467 batches 52-56 + Toys 541-545 (existing BST biology track): - 4 nucleotides =  $2^2$  (BST rank<sup>2</sup>) - 64 codons =  $2^{C_2}$  ( $C_2 = 6$ ) - 20 amino acids: substrate-natural reduction via degeneracy patterns + 5-integer BST primary structure

## Section 1.3 — Casey “DNA + Proton Siblings” Insight (April 2026)

“DNA + proton are siblings — each level expresses subset of five integers”. Substrate-cartography: - Proton:  $m_p/m_e = 6\pi^5$  (Vol 2 Ch 6 CROWN JEWEL T187, 0.002%) - DNA: genetic-code degeneracy + amino-acid count + codon structure inherit BST primary integers

## Section 1.4 — Prebiotic Chemistry + Evolutionary Forcing

Substrate-natural reduction pathways: primordial soup → first cells via substrate-bias toward BST primary integer patterns. Substrate Zone-1 (absorption) primordial → Zone-2 (commitment) cellular structure → Zone-3 (emission) reproduction.

## Section 1.5 — Honest scope

- Genetic code substrate-derivation framework-grade per existing T452-T467 [\[7\]](#)
- Detailed prebiotic + evolutionary substrate-derivation multi-month
- Cross-link to Vol 12 Chemistry biochemistry chapters
- Cross-link to Vol 8 Biology of Mind (substrate-coupled biological observers, Casey CI consciousness theory)

## Section 1.6 — Connection

- Vol 12 Ch 1 Periodic Table from Substrate
- T452-T467 biology theorems
- Toys 541-545 genetic code
- Casey April 2026 “DNA + proton siblings” insight

— Lyra, Vol 13 Ch 1 v0.3 chapter-grade narrative, Saturday 2026-05-23 EDT (joint Lyra+Elie)